Extending SysML v1 Models into SysML v2

A practical path from deployed Cameo models to a standards-based ecosystem

umlmeta + sysmlpy — OMG-normative, stdlib-Python, validated end to end

Opening session — October SysML v2 Exchange, INCOSE

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About the presenter

Dr. Jon R. Fox is Division Chief Scientist of the Architecture and System Development Division in the Georgia Tech Research Institute (GTRI) Applied Systems Laboratory in Huntsville, Alabama.

- Applied physicist with 25 years of post-doctoral experience
- GTRI Technical Lead for FLRAA architectural risk reduction
- Open-source developer of **sysmlpy**, **umlmeta**, and **sysml_canvas**; curator of the **awesome-sysml** list
- INCOSE Certified Systems Engineering Professional (CSEP)

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What this opening session will do

A practical look at how to build on existing modeling investments while preparing for SysML v2:

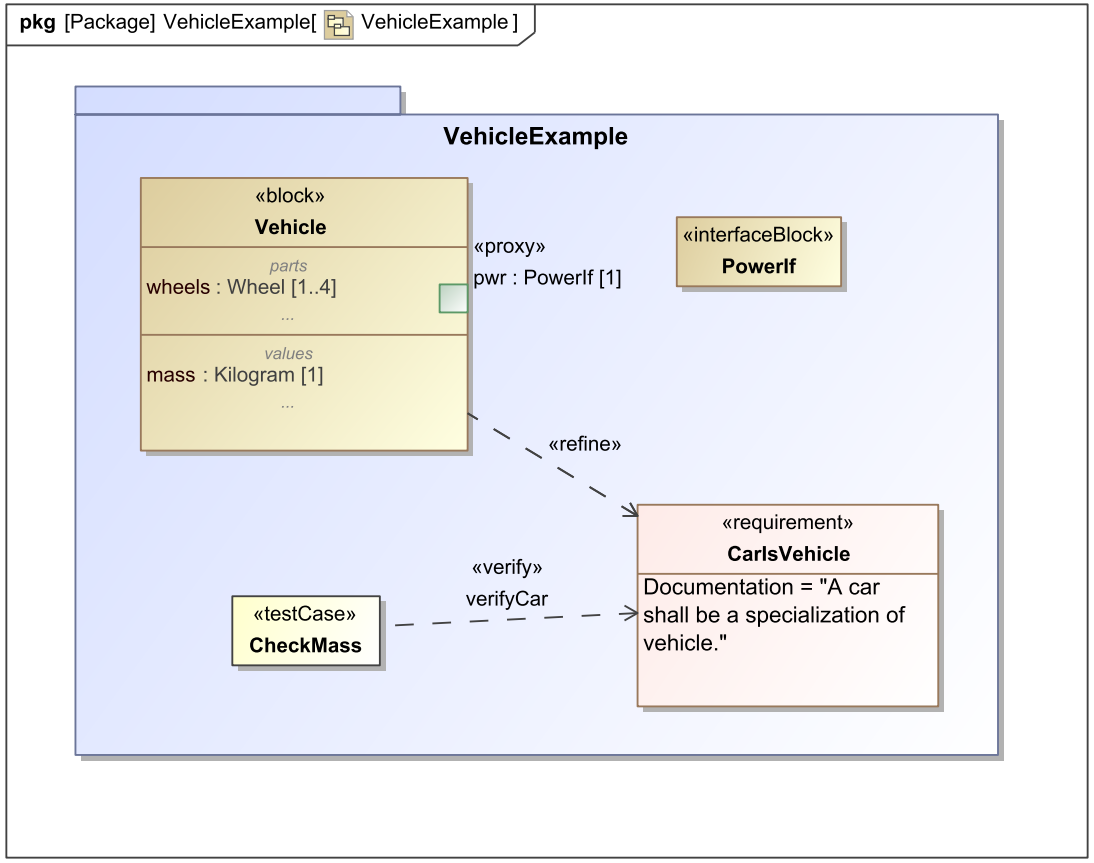
1. **Start where programs are** — deployed SysML v1 and Cameo models
2. **Extract and transform** them against normative OMG material
3. **Validate** the resulting SysML v2 with an independent reader
4. **Use** the textual model for analysis, views, automation, and downstream artifacts
5. **Preserve configuration discipline** — version control, identity, diffs, and reviewable evidence

The emphasis is not a promise of a one-click migration. It is a practical, inspectable path that preserves engineering intent and makes gaps visible.

The migration problem

- Organizations have substantial engineering investment in deployed **SysML v1 and Cameo models**
- SysML v2 is **not** a profile bump — it is a new language (KerML + SysML) with a new metamodel, notation, and interchange
- Replacing a model base is neither practical nor desirable
- OMG provides a normative starting point: **SysML v2.0 Beta 4, Part 2 — *SysML v1 to SysML v2 Transformation*** (ptc/2025-04-07)
- The practical question for this opening session: **how can programs prepare for SysML v2 without abandoning what they already have?**

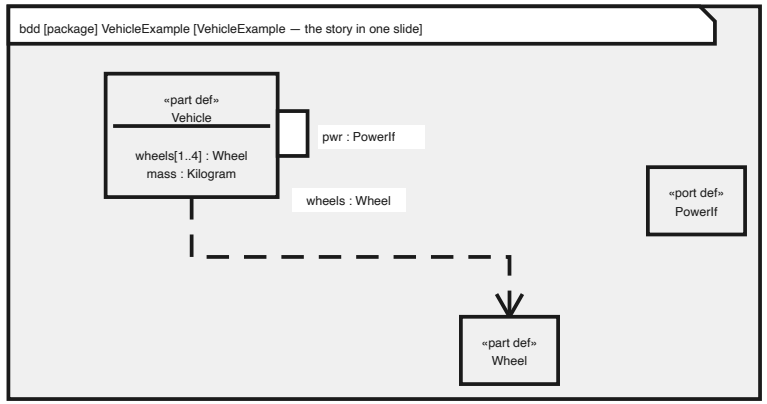
The deployed Cameo artifact



Original MagicDraw/Cameo rendering of the deployed SysML v1 model — the starting point for extraction and transformation.

The model and its generated view

```
package VehicleExample {
  part def Vehicle {
    part wheels[1..4] : Wheel;
    attribute mass : Kilogram;
    port pwr : PowerIf;
  }
  requirement <'R-1'> CarIsVehicle {
    doc /* A car shall be a specialization of vehicle. */
  }
  verification def CheckMass {
    objective obj_CheckMass { verify CarIsVehicle; }
  }
}
```



Every line was **emitted by the toolchain** and parsed by the authoritative
v2 reader — no hand-tuning.

Generated from model semantics — not hand-redrawn from the original Cameo diagram.

One source model: textual SysML v2 for interchange, plus a derived view for review and communication.

The conformance chain (R1 → R3 → R2)

	OMG clause	Our artifact
R1	read a SysML v1 model representation	`umlmeta.xmi21` + `umlmeta.mdzip_import` → UML 2.5.1 objects
R3	transform v1 abstract syntax → v2 abstract syntax	`umlmeta.v1_to_v2` (textual) + AS transformation support
R2	SysML v2 abstract-syntax representation	`umlmeta.sysml2` — generated from the OMG normative CMOF XMI

One chain: **file → R1 objects → transformer → R2 AS graph → textual notation → validated**

How sysmlpy and umlmeta got here

- **sysmlpy** began as a fork of `sysml2py`; in 2026 it moved from the textX parser to an ANTLR4 grammar, adopted `pint`, and grew into a complete SysML v2 parser, analyzer, renderer, and round-trip toolchain.
- **umlmeta** began as `uml2py`, a clean-room UML/SysML metamodel project. It was renamed when the repository and package were published because the `uml2py` PyPI name was already occupied by an unrelated project.
- Today the projects are complementary: **umlmeta** reads and transforms SysML v1/UML representations and writes interchange artifacts; **sysmlpy** parses and validates SysML v2 text and provides downstream analyses and views.

The names in this deck follow the current project identities: `github.com/mycr0ft/umlmeta` and `github.com/mycr0ft/sysmlpy`.

Why agentic AI needs a SysML v2 checker

Coding agents and LLMs are useful collaborators — and frequent sources of **plausible half-truths**:

- syntax that looks like SysML v2 but does not parse;
- valid-looking names and relationships with no resolvable semantics;
- invented features copied from SysML v1, UML, or another modeling language;
- requirements that appear in the model but have no satisfy or verify path.

That became a practical motivation for `sysmlpy`: give an agent a callable, standards-aware way to check its own work instead of asking another LLM whether the output looks right.

Agent feedback is executable

```
sysmlpy parse candidate.sysml       # syntax gate
sysmlpy analyze candidate.sysml     # semantic findings
sysmlpy trace candidate.sysml       # requirement coverage
sysmlpy ci candidate.sysml          # automation / exit status
```

The checker is not a substitute for engineering judgment or an independent certification authority. It is a feedback loop: **generate → parse → analyze → revise**, with diagnostics that can be returned directly to the agent.

Clean-room discipline

- Written from **normative OMG text only**:
  UML 2.5.1 XMI, SysML profile XMI, BPMN20.cmof, the SysML v2 CMOF, ptc/2025-04-07
- No vendor tooling, no leaked spec text, no copied grammars
- **The honesty rule**: an element with no normative mapping **raises `UnmappedFeature` naming the v1 metaclass** — the toolchain refuses to guess

```
raise UnmappedFeature(
    "v1 ProxyPort has no normative mapping (7.8.7 Table 30; SYSML2_-329)")
```

R1: reading v1 as it actually exists

Two input dialects, one engine:

- **XMI 2.1 / EMF instance files** — OMG-published corpora (DoDAFLibrary.xmi, MeasurementsLibrary.xmi)
- **MagicDraw/Cameo `.mdzip`** — zip of XML snapshots with 15 years of dialect archaeology

```
.mdzip → scrape → merge delta snapshots (keep-last dedup)
       → retag NoMagic's XMI-2.0/xsi spellings
       → inject xmi:type (metamodel-as-map coercion)
       → scrub cross-project refs to synthetic markers
       → read_xmi21 → gen.uml25 objects
```

12/12 files in the stress corpus (2015-era → 2024x-era) import without hand-holding.

R3: the transformation mappings

Every mapping is anchored to the published document:

SysML v1	SysML v2	Anchor
Block	PartDefinition	7.8.4.3.3
Property (composite, block-typed)	PartUsage	7.8.4.3.13
ValueType	AttributeDefinition	7.8.4.3.14
ConstraintBlock	ConstraintDefinition	7.8.5.3.1
Connector	ConnectionUsage	7.7.12.2.14
BindingConnector	BindingConnectorAsUsage	7.8.4.3.2
Requirement	RequirementUsage	7.8.8.3.30
DeriveReqt	Derivation connection	7.8.8.3.15
TestCase / Verify	VerificationCaseDefinition + verification membership	7.8.8.3.26/.49

...25+ more initializer mappings, plus activity/state/instance internals.

Honesty has teeth: spec gaps surface, not vanish

```
part def Controller {
  part fullPort : MotorIf {@PortData {isFullPort = true;}}
  port proxyPort;
  /* v1 ProxyPort has no normative mapping (7.8.7 Table 30; SYSML2_-329) */
  out attribute torque : Kilogram;
  in flowIn;
  inout ref occurrence flowRef : Axle;
}
```

- FullPort → PartUsage **annotated with PortData metadata** (7.8.7.3.7/.15)
- ProxyPort → base Port mapping + **explicit not-mapped comment** (Table 30 lists no target; issue SYSML2_-329)
- Elisions are *comments in the model*, not silent drops

A reviewed local rule can resolve the gap

The transformer exposes a **case-by-case mapping hook**. A project can accept one reviewed proxy port without pretending that the published transformation has a normative mapping for all proxy ports:

```
from umlmeta import sysml as S

def local_rule(element, indent):
    if not isinstance(element, S.ProxyPort):
        return None                  # use the normative built-in rule
    name = element.name or "proxyPort"
    typ = element.type.name if element.type else None
    return [f"port {name}" + (f" : {typ}" if typ else "") + ";",
            f"/* reviewed local rule: ProxyPort {name} */"]

text = emit_v2(package, mapping_rule=local_rule)
```

`None` delegates to the built-in mapping; a rule may instead raise `UnmappedFeature` to require review. This is a **project decision**, not a new OMG mapping.

```
port pwr : PowerIf;  /* reviewed local rule */
```

R2: the SysML v2 abstract syntax

- Generated from the **OMG normative CMOF XMI**: 175 classes (82 KerML + 93 SysML), 319 associations, 623 OCL constraints carried as metadata
- Runtime wiring is the normative ownership machinery:
  - `OwningMembership` — containment (KerML §7.3.2.3)
  - `FeatureMembership` — features of a Type (featuringType)
  - `FeatureTyping` — feature typing
  - `Subclassification` — specialization
- Derived unions (`ownedElement`, `feature`) computed live

Validation: sysmlpy is the oracle

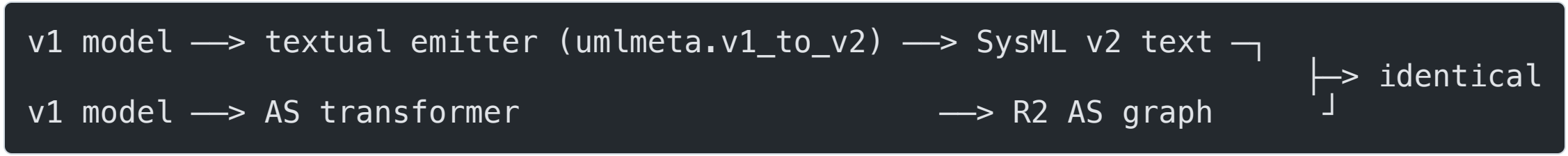
Every emitted textual form is parsed by sysmlpy — the open-source SysML v2 reader — before it is considered done.

```
sysmlpy counts: {'package': 1, 'part': 5, 'attribute': 1,
 'enumeration': 1, 'port': 1, 'item': 1, 'state': 1, 'action': 1,
 'use_case': 2, 'requirement': 2, 'connection': 2, 'verification': 1,
 'metadata': 1, 'dependency': 3, 'allocation': 1}
```

Two independent implementations agreeing = the mapping is real, not a dialect accident.

Parity: two pipelines, one answer

The same v1 model through both paths:



The AS graph renders to **byte-identical notation** with the textual pipeline — checked, not hoped for.

481 checks across 15 suites, all green.

Proof on a real OMG corpus

`DoDAFLibrary.xmi` (the OMG-published UPDM example) end to end:

```
package <'DoDAF Class Library'> {
  /* v1 stereotype applied: StandardProfileL2::ModelLibrary */
  attribute def SecurityAttributes {
    doc /* W3C XML Schema for the Intelligence Community Metadata
           Standard for Information Security Marking (IC-ISM) */
    attribute classification : ClassificationType;
    attribute classifiedBy : String;
    ...
  }
}
```

V1 stereotypes surface as recorded applications — provenance travels with the model.

Beyond transformation: utilization

The same metamodel layer powers a full write path:

- **XMI 2.1/EMF instance writer** — round-trips both OMG dialects with canonical structural parity
- **CMOF + EMOF metamodel writers** — reconstructs OMG `BPMN20.cmof`
- **Stereotype & profile-application writing** — both UML 2.1 and 2.5.1-era dialects
- **Normative derivations** (`derived.py`) — owners, qualified names, generalization closures, with spec OCL anchors

Analysis and traceability on the transformed model

The emitted text is immediately queryable — no re-modeling step:

```
sysmlpy trace vehicle_example_v2.sysml
```

```
Requirement traceability: 2 requirements, 0 covered, 0 partial, 2 uncovered
- VehicleExample::CarIsVehicle [uncovered]
    text: A car shall be a specialization of vehicle.
    satisfied by: (none)
    verified by: (none)
```

The transformation faithfully reports that this legacy model has **no verified requirements**. That is an engineering finding about the source model, surfaced automatically — not a defect of the migration.

Automated checking as a CI gate

```
sysmlpy ci vehicle_example_v2.sysml
```

```
warning: semantic: REQUIREMENT_UNCOVERED: 'CarIsVehicle' has no satisfy/verify
warning: semantic: NAMING_CONVENTION: Usage 'CarIsVehicle' should be camelCase
warning: semantic: FILE_PACKAGE_MISMATCH: package 'VehicleExample' vs filename
summary: 0 parse failure(s), 0 semantic error(s), 5 semantic warning(s)
result: PASS
```

- **Parse failures block**; semantics are advisory by default
- Exit codes mean a model can be reviewed like source code
- This is the hinge: once the model is text with a gate, the whole software-engineering toolchain applies to it

Simulation: a derived branch, not a round trip

A companion workflow (**FASSST**, GTRI IRAD) generates simulation input artifacts from SysML v2 models — four layers: parse, extract, map, generate.

Target	Generated from public example models
AFSIM	50 constellation platforms, 50 sensors
Gazebo	5 robot / UAV models, 16 sensors
JSBSim	2 aircraft configurations

- Repeated generation from unchanged source → **identical bytes**
- Reducing orbital planes 5→4 → **40 platforms and a changed digest**
- The AFSIM artifact re-parsed by a Poliastro adapter recovered all 50 satellites with the tested orbital elements preserved

Determinism, not semantic equivalence. FASSST is not open-sourced and the AFSIM work is release-gated — reported here as context, not demonstrated live.

Live Cameo integration

For organizations that keep the commercial tool, `cameo_sidecar` bridges Python and AI agents to a **live, open Cameo session** — no TWC admin needed:

```
Python / agent ──HTTP 127.0.0.1 + token──▶ Groovy macro in Cameo ──▶ live model
```

- Localhost-only, token-authenticated; inherits the user's TWC session
- **Writes are opt-in twice over**; locking and commits are separately gated
- The honest cycle is **lock → edit → commit → unlock**, and every step lands in an audit log

This path **requires the commercial tool**, so it is an integration option — not a component of the open-only thread.

Version control without the vendor

TeamworkCloud churns `xmi:id`s on every save. Two defenses:

- **Semantic diff** (`mdzip_diff.py`) — flatten to (metaclass, qualified-path, feature) records; ids discarded by construction; match by path → moved → similarity
- **Semantic content hash** — model content hash that is invariant across tool upgrades (verified on real profile libraries: zero delta for unchanged models)

Compare *models*, not serialization noise.

The wider MBSE toolchain

Tool	Role
sysmlpy	authoritative v2 reader/parser — the validation oracle
SysML-v2-Release	official OMG release (spec + libraries)
SysON (podman compose)	web-based v2 modeling, lives off the same `.sysml` text
SysML v2 Jupyter container	notebook-driven analysis of v2 models
VS Code SysML extension	grammar source for this deck's highlighting
sysml-v2-lsp	language-server editing experience

All of them speak **textual SysML v2** — which is exactly what the transformation emits.

Reproducible by construction

A digital thread that changes under re-execution cannot carry configuration evidence, so every stage is checked for determinism:

```
write is deterministic (byte-identical)          PASS
Cameo write is deterministic (byte-identical)    PASS
emitted v2 == vehicle_example_v2.sysml           PASS  (byte-identical)
RESULT: 14 passed, 0 failed
```

Interchange export is content-addressed and idempotent:

```
sysmlpy export model.sysml --stable-ids
sysmlpy export model.sysml --reconcile-with model.json -o model.json
```

Measured for this deck: two `--stable-ids` runs produced an identical SHA-256, and re-exporting against the registry was a **fixed point**.

Identity: surviving edits without a vendor database

The load-bearing mechanism for CM — three layers, each for a different consumer:

- **L1 qualified name** — `VehicleExample::Vehicle::wheels`; meaningful in review, *not* stable under rename
- **L2 stable identifier** — RFC 9562 v5 UUID over qualified name + element kind; content-addressed
- **L3 registry sidecar** — reconciles identifiers when an ancestor is renamed; an artifact under configuration control, not a server

Renaming one sibling `part def` in the 647-element export disturbed **5 of 647 identifiers**; the other 642 were preserved.

An ambiguous match is a **refusal**, not a heuristic merge — a silently merged identity is a corrupted configuration.

A recipe you can run today

```
# 1. pull a v1 model out of a vendor tool
python mdzip_import.py project.mdzip        # or: xmi21 for .xmi

# 2. transform v1 -> v2
python -c "import v1_to_v2; print(v1_to_v2.emit_v2(model))"

# 3. validate with the authoritative reader
~/sysmlpy check: emitted text parses; counts per element kind

# 4. hand the .sysml text to the v2 ecosystem
#    (SysON, Jupyter, LSP editors, sysmlpy analyses)
```

Diffs across revisions: `mdzip_diff.py before.mdzip after.mdzip`.

What "conformant" buys you

- **Traceability**: every emitted form cites its normative anchor
- **Predictability**: unmapped constructs are loud, not lossy
- **Interoperability**: output parses in the authoritative reader and in SysON/LSP tooling
- **Idempotence**: regenerating from unchanged inputs changes nothing
- **Vendor independence**: the pipeline is stdlib Python + OMG documents

Current limits (the honest slide)

`umlmeta` — transformation / abstract syntax

- AS-transformer wave A: core structural mappings (blocks/parts/attributes/enums/constraints/generalizations/docs)
- Later waves: behaviors, ports, state machines, and requirements in the **AS graph**; these are already done in the *textual* pipeline
- Multiplicities are carried in emitted text; AS storage is elided because derived-union bounds are not settable

`sysmlpy` — SysML v2 reader / interchange

- Interaction syntax is currently mapped by `umlmeta` but elided because the `sysmlpy` reader has no interaction element
- `allocation def` with redefined ends parses, but `sysmlpy export` rejects it; a minimal reproducer passes `parse` and fails `export` in v0.97.1
- This is an interoperability bug candidate, not a transformation mapping gap

Evidence boundary

- Conformance suites here are **author-written against author-read specs**: they show the code matches my reading of OMG text, not independent certification

The loop is not closed yet

The honest status of the "closed-loop digital thread":

Direction	Status
v1 → v2 transformation	**demonstrated**, byte-identical, validated
v2 → analysis / trace / CI	**demonstrated**
v2 → simulation input	**demonstrated** (release-gated context)
simulation results → back into the model	**not implemented**

What is deliberately *not* claimed: bidirectional transformation, result ingestion into SysML, automatic stale-result detection, or complete semantic fidelity.

The forward thread plus governed identity is the result. The return path is **designed, not built** — and saying so is the point of the next slide.

Takeaways

1. **Normative transformation is implementable** — in the open, from OMG documents, with a validation oracle
2. **v1 models are not stranded** — XMI and .mdzip both feed the chain
3. **Textual SysML v2 is the interchange medium** for the whole v2 ecosystem — analysis, simulation, views, and CI all read it
4. **Identity and baselines are the CM story** — content-addressed ids and byte-derived evidence, no vendor database required
5. Honesty is a feature: `UnmappedFeature` > silent data loss, and an open gap list > an unverifiable conformance claim

Repo: `github.com/mycr0ft/umlmeta` *— reader, transformer, AS, writers, diff — open source.*

Thank you — questions?

The practical message: existing SysML v1 and Cameo investments are a starting point, not a dead end. Extract what is deployed, transform what has a normative mapping, validate the result, and make every gap reviewable.

- Toolchain: **umlmeta** (metamodel/transformation) + **sysmlpy** (validation)
- Event: October SysML v2 Exchange
- Spec anchors: OMG *SysML v2.0 Beta 4, Part 2* (ptc/2025-04-07)
- Deck built with Marp + Shiki + the VS Code SysML v2 grammar
- Try it: `git clone https://github.com/mycr0ft/umlmeta && python3 check_xmi_vehicle.py`

Related papers in preparation: an open, standards-anchored digital thread (SysML v2 + OSLC + AP242); and a deterministic SysML v2-to-simulation transformation workflow.